Inequalities — problem 7.29

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Let $a_i \geq 1$ for $i = 1, \ldots, n$. Show that

$$(1 + a_1)(1 + a_2) \cdots (1 + a_n) \geq \frac{2^n}{n + 1}(1 + a_1 + a_2 + \cdots + a_n).$$